

Announcements

- ▶ Grading: I will catch up soon. I'm sorry!
- ▶ Happy (perfect square)³ day! (and nice pythagoras day!)
(in this perfect-square year!)
- ▶ Schedule:
 - Today (Lecture)**: Electromagnetic Spectrum (light) and Blackbody radiation
 - Tomorrow (Lab)**: "all the light you cannot see"
 - Thursday (Discussion)**: Blackbody radiation and estimating Earth's Temperature
- ▶ Next Week: Greenhouse Effect I

Today's Reading Quiz (3 min)

On your own sheet of paper. . .

How does the Sun's energy get to the Earth? How would you describe that energy/energy-transfer, and what are other examples (not the Sun, but same type of energy transfer)?

Turn in to Blackboard as scanned pdf (or image is fine).

Review: Energy Types, Energy Transfer, Power

Energy Types (units of Joules = "J")

- ▶ "External": Kinetic (K) of object, Potential (V) in gravitational or spring "field"
- ▶ "Internal": Thermal or mechanical kinetic/potential of atoms or internal parts; Chemical; Nuclear
- ▶ (new today) "Electromagnetic": carried by photons/light

Energy Transfer Mechanisms (units: J)

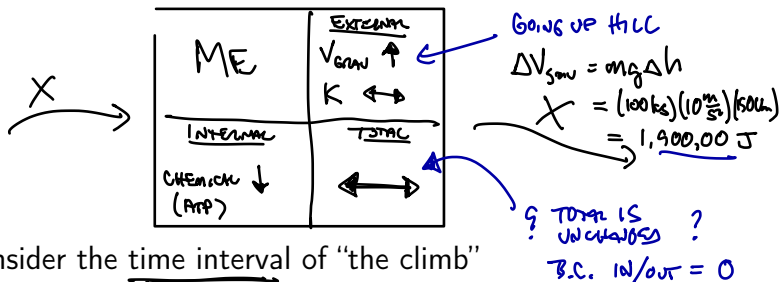
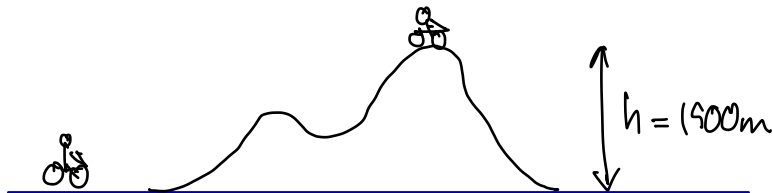
- ▶ Mechanical Work: $W = F \times \Delta x$ (force over a distance)
- ▶ Heat Conduction: Objects in contact (high T to low)
- ▶ Convection: fluid motion of internal parts
- ▶ (today) Radiation: transmission of photons

Power: Energy per time ($P = \frac{E}{\Delta t}$, in units of $\frac{J}{s}$)

Flux: Power per area ($F = \frac{E}{\Delta t \times A}$, in units of $\frac{J}{s \times m^2}$)

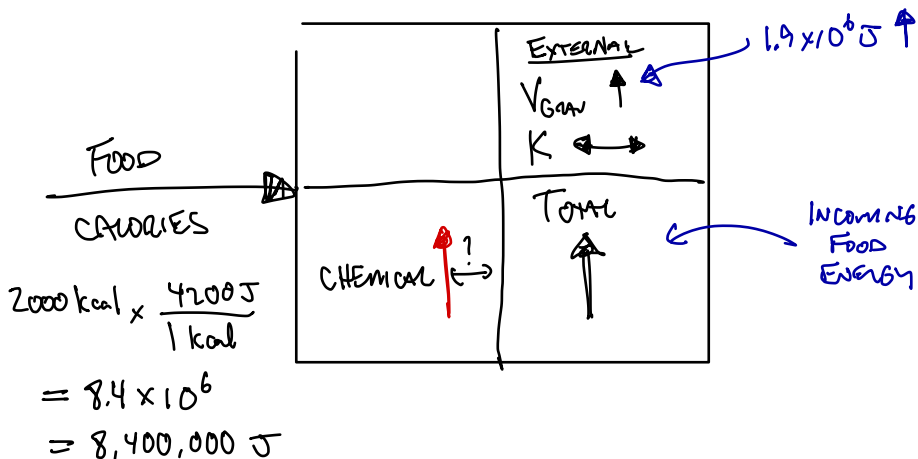
One last good day for the professor

Let's sketch an energy flow diagram for the professor climbing Mount Ventoux ($h = 1900$ m).



First consider the time interval of "the climb"

One last good day for the professor (part 2)



Now consider the time interval of "the climb" and "the previous day's meals" (2000 calories = 2000 kcal; 1 kcal = 4200 J)

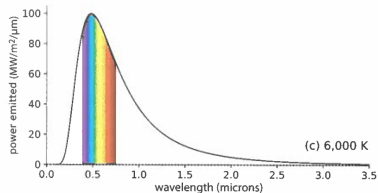
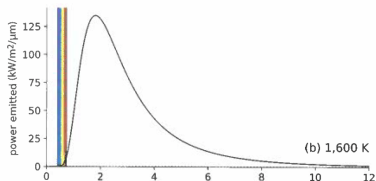
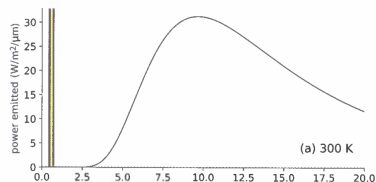
One last good day for the professor (part 3)

Q: Where is all of the food energy going?

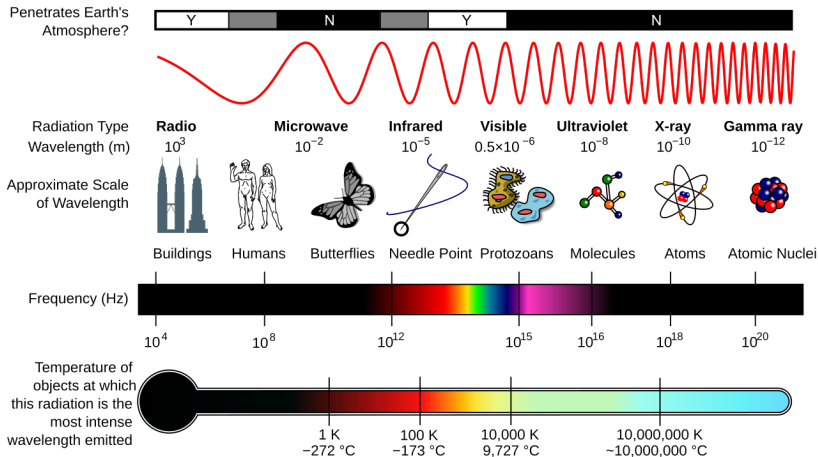
A: Turns out the professor is a light bulb.

Any object with a temperature ($< \infty$ K) emits electromagnetic radiation

(There is also similar heat losses due to convection/conduction.)



Electromagnetic Radiation (light!)



Equations for Blackbody Radiation

Wien's Law

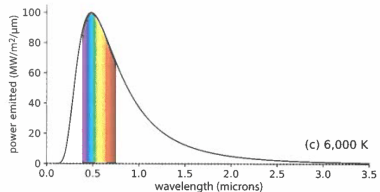
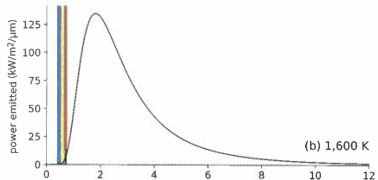
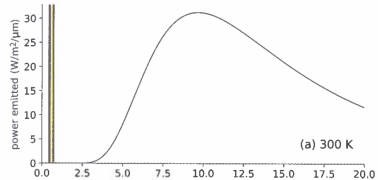
$$\lambda_{\text{peak}} = \frac{2900 \text{ } \mu\text{m} \cdot \text{K}}{T}$$

Stefan-Boltz. Law (total flux)

$$F = \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}} \right) T^4$$

Peak "Spectral Flux"

$$F_{\lambda} = \frac{F}{1.5 \times \lambda_{\text{peak}}}$$



Graphing Bb Radiation

$$\lambda_{\text{peak}}(\text{Human}) = \frac{2900 \mu\text{m} \cdot \text{K}}{310 \text{ K}} \approx 10 \mu\text{m}$$

$$F(\text{Human}) = (6 \times 10^{-8} \dots) (310 \text{ K})^4 = 554 \text{ J/m}^2/\text{s}$$

Let's graph (on the log-log paper provided) the Bb radiation of three objects:

- ▶ Sun ($T = 6000 \text{ K}$)
- ▶ Human ($T = 310 \text{ K}$)
- ▶ Liquid Nitrogen ($T = 77 \text{ K}$)

$$E_{\text{tot}} = F \times A \times \Delta t \quad \text{1 day}$$

$$= (55 \dots 2 \text{ m}^2) (60 \times 60 \times 24)$$

$$= 96,000,000 \text{ J}$$

We use Wien's law and the "spectral flux" peak, as well as a 10% width of $(\lambda/2, \lambda \times 3)$.

$$F_{\text{NET}} = (\quad) (T_{\text{Human}}^4 - T_{\text{Environment}}^4)$$

$$\Rightarrow 100 \text{ W} = 100 \frac{\text{J}}{\text{s}} \Rightarrow$$